Evolution of dengue in Sri Lanka-changes in the virus, vector, and climate.

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Despite the presence of dengue in Sri Lanka since the early 1960s, dengue has become a major public health issue, with a high morbidity and mortality. Aedes aegypti and Aedes albopictus are the vectors responsible for the transmission of dengue viruses (DENV). The four DENV serotypes (1, 2, 3, and 4) have been co-circulating in Sri Lanka for more than 30 years. The new genotype of DENV-1 has replaced an old genotype, and new clades of DENV-3 genotype III have replaced older clades. The emergence of new clades of DENV-3 in the recent past coincided with an abrupt increase in the number of dengue fever (DF)/dengue hemorrhagic fever (DHF) cases, implicating this serotype in severe epidemics. Climatic factors play a pivotal role in the epidemiological pattern of DF/DHF in terms of the number of cases, severity of illness, shifts in affected age groups, and the expansion of spread from urban to rural areas. There is a regular incidence of DF/DHF throughout the year, with the highest incidence during the rainy months. To reduce the morbidity and mortality associated with DF/DHF, it is important to implement effective vector control programs in the country. The economic impact of DF/DHF results from the expenditure on DF/DHF critical care units in several hospitals and the cost of case management.

Phylogeography and spatio-temporal genetic variation of Aedes aegypti (Diptera: Culicidae) populations in the Florida Keys.

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Aedes aegypti (L.) is the principal mosquito vector of dengue fever, the second-most deadly vector-borne disease in the world. In Ae. aegypti and other arthropod disease vectors, genetic markers can be used to inform us about processes relevant to disease spread, such as movement of the vectors across space and the temporal stability of vector populations. In late 2009, 27 locally acquired cases of dengue fever were reported in Key West, FL. The last dengue outbreak in the region occurred in 1934. In this study, we used 12 microsatellite loci to examine the genetic structure of 10 Ae. aegypti populations from throughout the Florida Keys and Miami to assess gene flow along the region's main roadway, the Overseas Highway. We also assessed temporal genetic stability of populations in Key West to determine whether the recent outbreak could have been the result of a new introduction of mosquitoes. Though a small amount of geographic genetic structure was detected, our results showed high overall genetic similarity among Ae. aegypti populations sampled in southeastern Florida. No temporal genetic signal was detected in Key West populations collected before and after the outbreak. Consequently, there is potential for dengue transmission across southeastern Florida; renewed mosquito control and surveillance measures should be taken.

Transcriptomics and disease vector control.

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Next-generation sequencing can be used to compare transcriptomes under different conditions. A study in BMC Genomics applies this approach to investigating the effects of exposure to a range of xenobiotics on changes in gene expression in the larvae of Aedes aegypti, the mosquito vector of dengue fever.

[Dengue: a growing risk to travellers to tropical and sub-tropical regions].

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Dengue is currently the most common arboviral infection worldwide. Due to global climate change and other factors, the vector of the virus - the Aedes mosquito - has spread considerably over the past decades. Dengue is endemic in almost all tropical and sub-tropical regions of the world; meaning that approximately 40% of the world population is at risk of acquiring a dengue infection. The clinical features of dengue vary from a non-specific febrile illness (dengue fever) to at times fatal serious conditions such as dengue haemorrhagic fever (DHF) and dengue shock syndrome (DSS). Considering the limited possibilities of prevention it is anticipated that the incidence of dengue will increase in the future. It is expected that health-care providers in non-endemic regions will encounter dengue-infected patients with increasing frequency in their practices.

Aedes aegypti in Ho Chi Minh City (Viet Nam): susceptibility to dengue 2 virus and genetic differentiation.

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Aedes aegypti is the principal vector of dengue viruses, responsible for a viral infection that has become a major public health concern in Asia. In Viet Nam, dengue haemorrhagic fever was first detected in the 1960s and is now a leading cause of death in childhood. We studied the variability in competence of Ae. aegypti as a vector for dengue 2 virus and genetic differentiation in this mosquito species. Twenty mosquito samples collected in 1998 in Ho Chi Minh City were subjected to oral infection and isoenzyme polymorphism analysis by starch gel electrophoresis. Ae. aegypti populations from the centre of Ho Chi Minh City were genetically differentiated and their infection rates differed from those of populations from the commuter belt. These results have implications for insecticidal control during dengue outbreaks.

Detection and Serotyping of Dengue Viruses in Aedes aegypti and Aedes albopictus (Diptera: Culicidae) Collected in Surabaya, Indonesia from 2008 to 2015.

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Aedes aegypti and Aedes albopictus are the primary and secondary vectors, respectively, of dengue, the most important arboviral disease in the world. The aim of this study was to detect and serotype dengue viruses (DENV) in the vectors Ae. aegypti and Ae. albopictus in Surabaya, Indonesia. Between 2008 and 2015, 16,605 Aedes mosquitoes were collected in 15 sub-districts of Surabaya. Ae. aegypti was dominant (90.9%), whereas few Ae. albopictus were collected (9.1%). A total of 330 pools of adult Aedes mosquitoes were subjected to the serotyping of DENV by RT-PCR. DENV-1 (52.3%) was the most frequently detected serotype, followed by DENV-2 (40.3%), DENV-4 (4.6%), and DENV-3 (2.8%). The average minimum infection rate for Ae. aegypti in various sub-districts of Surabaya was 7.2 per 1,000 mosquitoes, while that for Ae. albopictus was 0.7 per 1,000 mosquitoes. The results showed that the predominantly circulating DENV serotype in mosquitoes continuously shifted from DENV-2 (2008) to DENV-1 (2009-2012), to DENV-2 again (2013-2014), and then back to DENV-1 (2015). The circulating

DENV serotypes in mosquitoes were generally consistent with those in humans. Therefore, the surveillance of infected mosquitoes with DENV might provide an early warning sign for the risk of future dengue outbreaks.

Comparative analysis of the susceptibility of Aedes aegypti and Japanese Aedes albopictus to all dengue virus serotypes.

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Dengue fever, caused by the dengue virus (DENV), is the most common viral infection transmitted by Aedes mosquitoes (mainly Ae. aegypti and Ae. albopictus) worldwide. Aedes aegypti is not currently established in Japan, and Ae. albopictus is the primary vector mosquito for DENV in the country, but knowledge of its viral susceptibility is limited. Therefore, we aimed to clarify the status of DENV susceptibility by comparing the infection and dissemination dynamics of Japanese Ae. albopictus to all known DENV serotypes with those of Ae. aegypti. After propagation of each DENV serotype in Vero cells, the culture supernatants were mixed with defibrinated rabbit blood and adenosine triphosphate, and the mixture was artificially blood-sucked by two colonies of Ae. albopictus from Japan and one colony of Ae. aegypti from a dengue-endemic country (Vietnam). After 14 days of sucking, the mosquito body was divided into two parts (thorax/abdomen and head/wings/legs) and total RNA was extracted from each sample. DENV RNA was detected in these extracted RNA samples using a quantitative RT-PCR method specific for each DENV serotype, and infection and dissemination rates were analyzed. The Japanese Ae. albopictus colonies were susceptible to all DENV serotypes. Its infection and dissemination rates were significantly lower than those of Ae. aegypti. However, the number of DENV RNA copies in Ae. albopictus was almost not significantly different from that in Ae. aegypti. Furthermore, Japanese Ae. albopictus differed widely in their susceptibility to each DENV serotype. In Japanese Ae. albopictus, once DENV overcame the midgut infection barrier, the efficiency of subsequent propagation and dissemination of the virus in the mosquito body was comparable to that of Ae. aegypti. Based on the results of this study and previous dengue outbreak trends, Ae. albopictus is predicted to be highly compatible with DENV-1, suggesting that this serotype poses a high risk for future epidemics in Japan.

Prohemocytes are the main cells infected by dengue virus in Aedes aegypti and Aedes albopictus.

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The primary disease vectors for dengue virus (DENV) transmission between humans are the mosquitoes Aedes aegypti and Aedes albopictus, with Ae. aegypti population size strongly correlated with DENV outbreaks. When a mosquito is infected with DENV, the virus migrates from the midgut to the salivary glands to complete the transmission cycle. How the virus crosses the hemocoel, resulting in systemic infection, is still unclear however. During viral infection and migration, the innate immune system is activated in defense. As part of cellular-mediated immunity, hemocytes are known to defend against bacteria and Plasmodium infection and may also participate in defending against DENV infection. Hemocytes are categorized into three cell types: prohemocytes, granulocytes, and oenocytoids. Here, we investigated which hemocytes can be infected by DENV and compare hemocyte infection between Ae. aegypti and Ae. albopictus. Hemocytes were collected from Ae. aegypti and Ae. albopictus mosquitoes that were intrathoracically infected with DENV2-GFP. The collected hemocytes were then identified via Giemsa staining and examined microscopically for morphological differences and viral infection. All three types of hemocytes were infected by DENV, though the predominantly infected cell type was prohemocytes. In Ae. aegypti, the highest and lowest infection rates at 7 days post infection occurred in prohemocytes and granulocytes, respectively. Prohemocytes were also the primary infection target of DENV in Ae. albopictus, with similar infection rates across the other two hemocyte groups. The ratios of hemocyte composition did not differ significantly between non-infected and infected mosquitoes for either species. In this study, we showed that prohemocytes were the major type of hemocyte infected by DENV in both Ae. aegypti and Ae. albopictus. The infection rate of prohemocytes in Ae. albopictus was lower than that in Ae. aegypti, which may explain why systemic DENV infection in Ae. albopictus is less efficient than in Ae. aegypti and why Ae. albopictus is less correlated to dengue fever outbreaks. Future work in understanding the mechanisms behind these phenomena may help reduce arbovirus infection prevalence.

Discriminable roles of Aedes aegypti and Aedes albopictus in establishment of dengue outbreaks in Taiwan.

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Aedes aegypti and Aedes albopictus were reported to be significant as vectors of dengue fever. In Taiwan, the latter is distributed throughout the island while the former appears only south of the Tropic of Cancer; i.e., 23.5°N. In the past decade, there were five outbreaks with over 1000 cases of dengue fever in Taiwan. Without exception, these outbreaks all occurred in the south where the two Aedes mosquitoes are sympartic. According to the Center for Disease Control of Taiwan, imported cases are thought to provide the seeds of dengue outbreaks every year. Mostly, the number of imported cases is greater in northern island, probably due to a larger population of travelers and imported workers from endemic countries. Looking at the example in 2002, northern, central, and southern parts of Taiwan reported 28, 11, and 13 imported cases, respectively. However, 54, 21, and 5309 total cases were confirmed in the corresponding regions over the entire year, indicating a significant skew of case distributions. A hypothesis is thus inspired that the existence of Ae. aegypti is a prerequisite to initiate a dengue outbreak, while participation of Ae. albopictus expands or maintains the scale until the de novo herd immunity reaches high level.

Molecular studies with Aedes (Stegomyia) aegypti (Linnaeus, 1762), mosquito transmitting the dengue virus.

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Dengue is an infectious viral disease, which can present a wide clinical picture, ranging from oligo or asymptomatic forms, to bleeding and shock, and can progress to death. The disease problem has increased in recent years, especially in urban and suburban areas of tropical and subtropical regions. There are five dengue viruses, called serotypes (DEN-1, DEN-2, DEN-3, DEN-4, and DEN-5), which belong to the Flaviviridae family and are transmitted to humans through infected mosquito bites, with the main vector the Aedes aegypti mosquito (Linnaeus, 1762). Studies performed with Ae. aegypti, aimed at their identification and analysis of their population structure, are fundamental to improve understanding of the epidemiology of dengue, as well for the definition of strategic actions that reduce the transmission of this disease. Therefore, considering the importance of such research to the development of programs to combat dengue, the present review considers the techniques used for the molecular identification, and evaluation of the genetic variability of Ae. aegypti.